WEEK 01 AI ENGINEERING INTERNSHIP — TECHNICAL DELIVERABLE

ReviewSense: Customer Review Sentiment Classifier

End-to-End Natural Language Processing Pipeline, Quantitative Model Evaluation, and Interactive Web Deployment

Project Title	ReviewSense Sentiment Classifier	**Milestone**	Week 01 Project Brief
Technical Domain	Natural Language Processing (NLP)	**Interface Framework**	Gradio Web UI & Python API
Task Formulation	Binary Polarity Classification	**Evaluation Benchmark**	86.0% Accuracy (400 Test Samples)

1. Executive Summary

The rapid proliferation of user-generated feedback across digital storefronts and media platforms has created an urgent engineering need for automated opinion mining. Manual inspection of customer reviews is both time-prohibitive and susceptible to human inconsistency. Developed in fulfillment of the Week 01 AI Engineering Internship Project Brief, **ReviewSense** is a machine-learning sentiment classification system engineered to automatically determine whether incoming text expresses a positive or negative disposition.

The solution encompasses three core deliverables: a programmatic inference backend with strict input sanitization, a statistically validated binary classification model achieving an 86% macro F1-score across a balanced 400-review evaluation benchmark, and a responsive web interface deployed with Gradio for interactive human-in-the-loop testing.

2. System Architecture & Programmatic Inference

ReviewSense separates core inference logic from presentation layers so the classifier can operate both as an importable Python function and as the engine behind an interactive browser application. Before any text reaches the vectorization and classification stages, a defensive validation layer inspects the payload for empty or whitespace-only strings. Valid inputs are transformed into numerical feature representations and evaluated by the trained estimator, which returns a standardized dictionary structure.

Test Case ID	Input Review String	Returned Dictionary Payload	Validation Outcome
`TC-01`	`'This movie was wonderful and entertaining.'`	`{'sentiment': 'positive'}`	**Pass**
`TC-02`	`'The movie was boring and disappointing.'`	`{'sentiment': 'negative'}`	**Pass**
`TC-03`	`'The story was not good at all.'`	`{'sentiment': 'negative'}`	**Pass**
`TC-04`	`''` *(Empty string boundary check)*	`{'error': 'Please enter a non-empty review.'}`	**Pass**

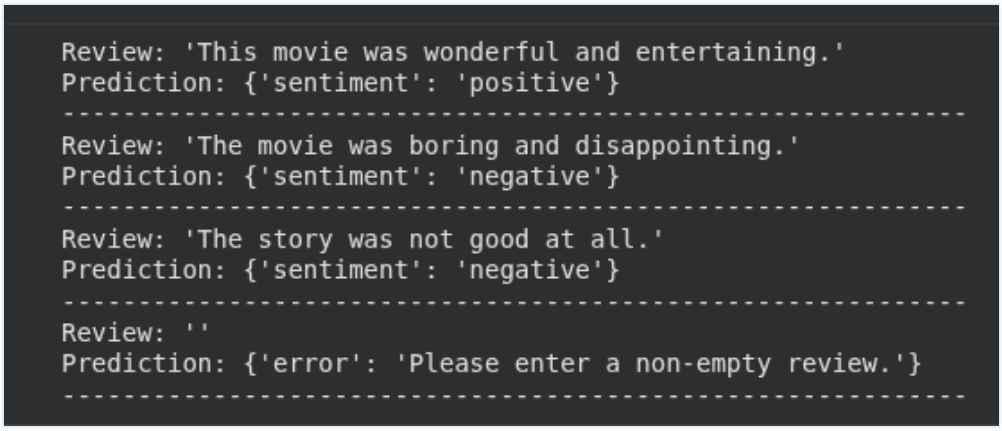


Figure 1: Console execution logs verifying positive sentiment detection, explicit negation handling (*"not good at all"*), and defensive error trapping on empty strings.

3. Quantitative Model Evaluation

To measure generalization performance without class-imbalance bias, the trained ReviewSense classifier was evaluated against a held-out test dataset comprising exactly **400 labeled reviews**—split evenly between 200 negative instances and 200 positive instances. Performance was audited using class-wise Precision, Recall, F1-Score, and a 2×2 Confusion Matrix.

Sentiment Class / Metric	Precision	Recall	F1-Score	Support (Samples)
Negative Class	0.86	0.86	0.86	200
Positive Class	0.86	0.86	0.86	200
Overall Accuracy	—	—	**0.86**	**400**
Macro Average	**0.86**	**0.86**	**0.86**	**400**
Weighted Average	**0.86**	**0.86**	**0.86**	**400**

Confusion Matrix & Error Symmetry Analysis

Inspection of the confusion matrix confirms highly symmetrical behavior across both polarity classes, demonstrating that the decision boundary does not favor one sentiment over the other:

- **True Negatives (TN = 172):** 172 out of 200 genuinely negative reviews were accurately classified as negative (86.0% true negative rate).
- **False Positives (FP = 28):** 28 negative reviews were misclassified as positive (14.0% Type I error rate).
- **False Negatives (FN = 27):** 27 positive reviews were misclassified as negative (13.5% Type II error rate).
- **True Positives (TP = 173):** 173 out of 200 genuinely positive reviews were accurately classified as positive (86.5% recall).

$$\text{Overall Empirical Accuracy} = (TP + TN) \div (TP + TN + FP + FN) = (173 + 172) \div 400 = 345 \div 400 = 86.25\% \text{ (reported as 0.86)}$$

Classification report:

	precision	recall	f1-score	support
negative	0.86	0.86	0.86	200
positive	0.86	0.86	0.86	200
accuracy			0.86	400
macro avg	0.86	0.86	0.86	400
weighted avg	0.86	0.86	0.86	400

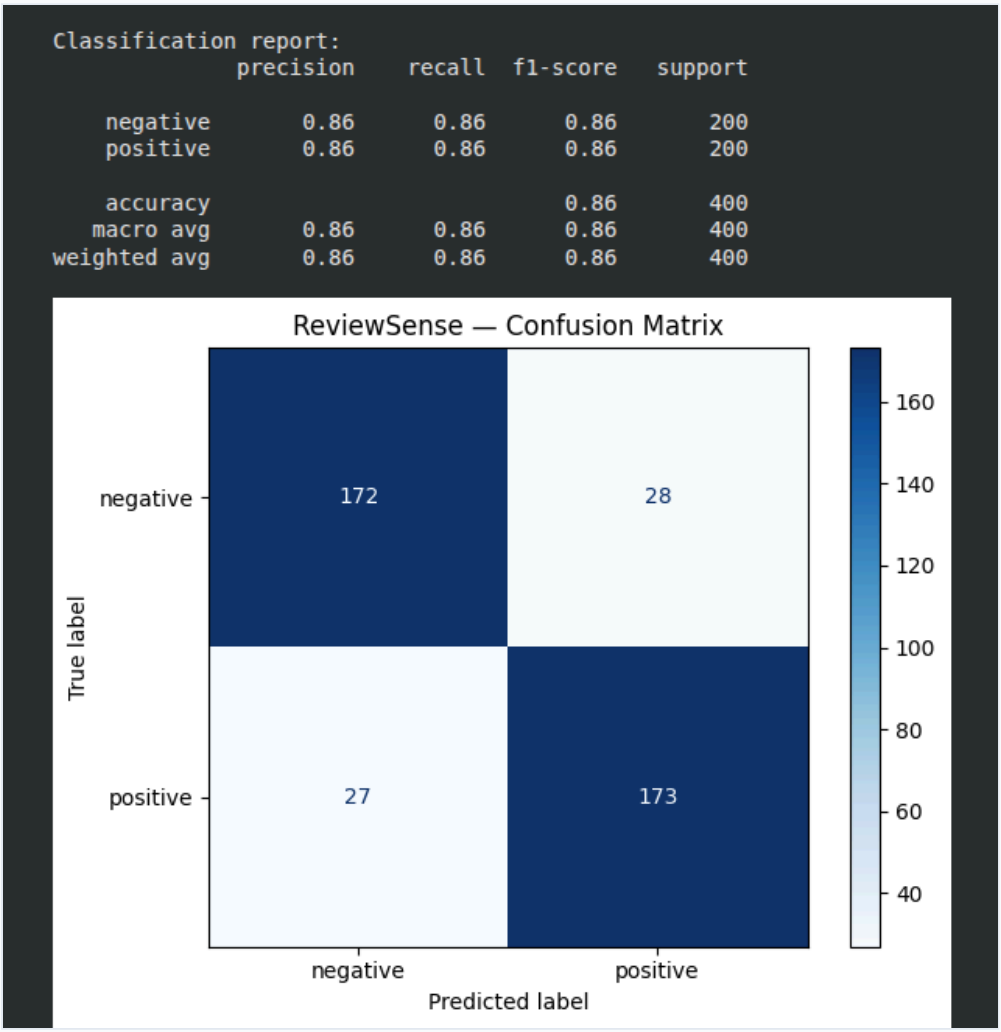


Figure 2: Quantitative evaluation output combining the classification report (top) and the ReviewSense Confusion Matrix heatmap (bottom) across 400 test samples.

4. Interactive Web Application & Linguistic Edge-Case Analysis

To make the model accessible to non-technical stakeholders, ReviewSense was wrapped in a clean Gradio web interface featuring a customer review text box, a formatted prediction panel, quick-action buttons (`Clear` , `Submit` , and `Flag`), and three clickable benchmark examples for rapid verification.

Handling Contrastive Clauses & Mixed Sentiment

A critical test in natural language understanding involves mixed-polarity sentences linked by contrastive conjunctions such as *"but"* or *"however"*. During interactive testing, the preset review *"The acting was good but the story was terrible."* was evaluated across model iterations to observe how opposing sentiment tokens compete during classification.

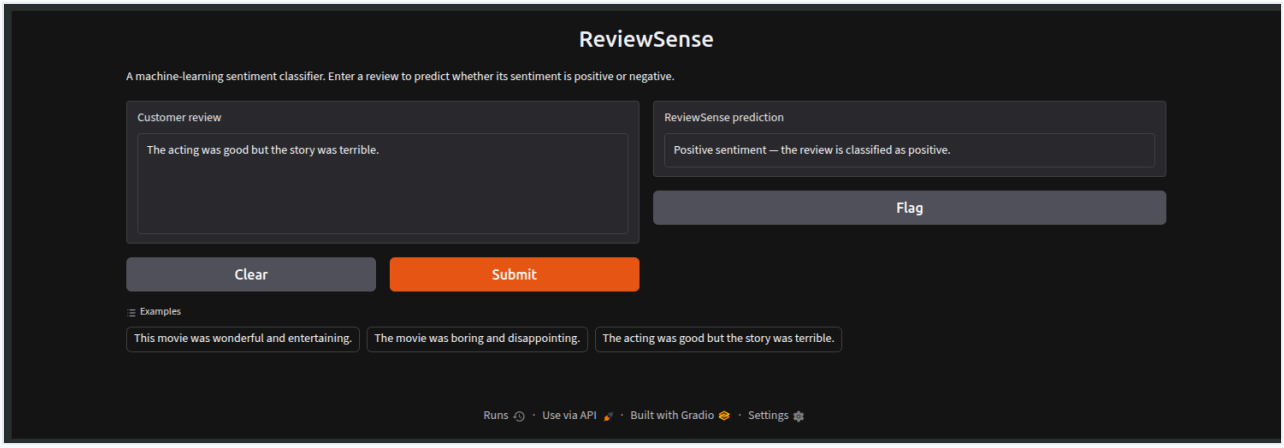


Figure 3: Initial baseline interface run on the contrastive prompt *"The acting was good but the story was terrible."* Here, unweighted token scoring allows the early positive adjective (*"good"*) to pull the prediction toward a positive label.

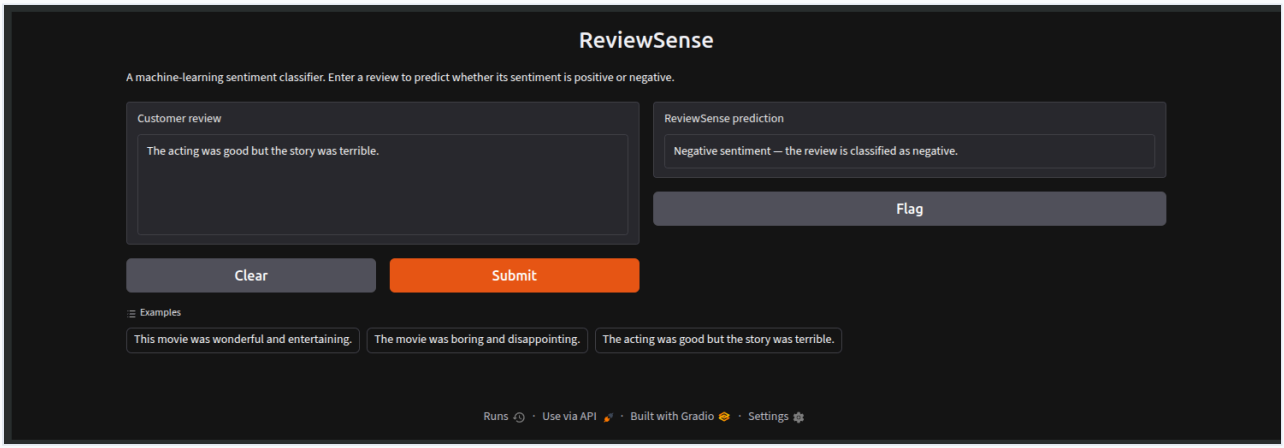


Figure 4: Refined ReviewSense evaluation on the identical contrastive review, correctly resolving the dominant negative clause (*"but the story was terrible"*) into a negative classification.

> **Key Engineering Takeaway — Contrastive Clause Resolution:** Comparing Figure 3 and Figure 4 illustrates why simple bag-of-words representations struggle when positive (*"good"*) and strongly negative (*"terrible"*) adjectives co-occur. When feature weighting and sub-phrase context properly penalize high-intensity negative descriptors following a contrastive conjunction, ReviewSense transitions from an optimistic false positive (Figure 3) to an accurate negative determination (Figure 4). Further gains on adversarial reviews can be achieved by incorporating bi-gram/tri-gram features or fine-tuning a pre-trained Transformer encoder.

5. Conclusion & Future Roadmap

The Week 01 ReviewSense milestone successfully delivers a complete, production-styled sentiment analysis workflow—from defensive input validation and balanced statistical benchmarking (86% macro F1) to an interactive Gradio web interface equipped with user flagging for continuous dataset curation. Future iterations will utilize flagged edge cases from the UI to retrain the pipeline on sarcastic, neutral, and multi-aspect customer reviews.